

Forecasting Rubric

Course: DS 5740-01 Advanced Stats for DS (2025F)

What Other Factors Might Influence the Time Series?	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
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What Other Factors Might Influence the Time Series?	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
You'll need to identify at least two other time series or variables that could impact the primary series. Think along the lines of economic indicators, weather data, or other relevant variables. Explain why you think these could be important and how you'll factor them into the analysis. These additional series will be incorporated throughout the analyses.	Two or more relevant time series were considered as part of the time series analyses	Two or more time series were considered as part of the time series analyses but one was not relevant	Only one other relevant time series were considered as part of the time series analyses	No additional factors were considered	/ 3

How Does the Time Series Look?	2 2 points	1 1 point	0 0 points	Criterion Score
Create some basic visualizations of both the primary time series and the other covariates you identified. Use line plots to spot trends, seasonality, and other features. Break the series down into its components — trend, seasonality, and noise. Discuss how your primary time series relates to other covariates and whether you suspect any patterns in the covariates relate to the patterns in the primary time series.	Visualizations were sufficient	More visualizations could have provided better inference for the analyses	Few or no relevant visualizations were provided	/ 2

How Good is the Model?	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
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How Good is the Model?	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
After you fit a model—including the covariates—check the residuals for any auto-correlation or signs that the model isn't capturing something important. Use plots and diagnostic tools to see how well it's doing. Specifically, comment on whether including the covariates improves the model.	All diagnostics were checked and appropriately interpreted	Some diagnostics were not check or some interpretations were missing or inappropriate	Some diagnostics were not check and some interpretations were missing or inappropriate	Diagnostics and interpretations were not provided	/ 3

Is the Model Robust Over Time?	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
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Is the Model Robust Over Time?	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
Test the model's stability over time, using techniques like cross-validation or splitting the dataset into training and test sets. Discuss how the covariates you included might impact the model's robustness or its ability to generalize to other datasets or periods. You should not use forecasts (yet) to evaluate how robust your time series are. Instead, you should be drawing conclusions based on the repro-	Cross-validation/data splitting was applied, covariates were discussed, and expected generalizability was interpreted.	One of the following was not provided: Cross-validation/data splitting was applied, covariates were discussed, and expected generalizability was interpreted.	Two of the following was not provided: Cross-validation/data splitting was applied, covariates were discussed, and expected generalizability was interpreted.	Cross-validation/data splitting was not applied, covariates were not discussed, and expected generalizability was not interpreted.	/ 3

Is the Model Robust Over Time?	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
ducibility of the parameters.					

Can You Make Forecasts?	2 2 points	1 1 point	0 0 points	Criterion Score
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Can You Make Forecasts?	2 2 points	1 1 point	0 0 points	Criterion Score
Generate at least one fore- cast that in- cludes both the time series of interest and the covariates. Compare your predictions to actual data (hold out some data like in 4. but actually make the fore- cast). Visualize the forecast, including con- fidence inter- vals, and ex- plain how much the addi- tional factors improved the forecast accu- racy using fit measures.	A forecast was created and a held out sample was used and addition- al factors were explained.	A forecast was created but a held out sample was not used or addi- tional factors were explained.	A forecast was not created	/ 2

Can We Try Other Models and Compare?	5 5 points	4 4 points	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
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Can We Try Other Models and Compare?	5 5 points	4 4 points	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
Try out at least one other model and compare its performance, both with and without the covariates. Use fit measures to evaluate each model and discuss the trade-offs. Which model works best, and how did the covariates impact the results?	Process was followed for both models with and without covariates		Components were missing for either the model with or without covariates		Components were missing for both models with and without covariates	Models were not compared	/ 5

Can We Try Other Models and Compare?	5 5 points	4 4 points	3 3 points	2 2 points	1 1 point	0 0 points	Criterion Score
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How Well Does the Model Generalize?	2 2 points	1 1 point	0 0 points	Criterion Score
Discuss how well your model, including covariates, will generalize to other similar datasets or future data. Point out any limitations and offer practical advice on how the model could be used in real-world situations. Highlight the role the covariates played in improving the model's performance.	A thorough discussion was provided about what model would be used and potential limitations. Practical and applied advice was provided for how the model could be used in the real-world.	The discussion provided could improve on what model would be used and potential limitations or the practical and applied advice was provided for how the model could be used in the real-world.	No discussion of the model usage or practical advice for the real-world was provided	/ 2

Total	/ 20
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Overall Score

A

18 points minimum

B

16 points minimum

C

14 points minimum

D

12 points minimum

F

0 points minimum